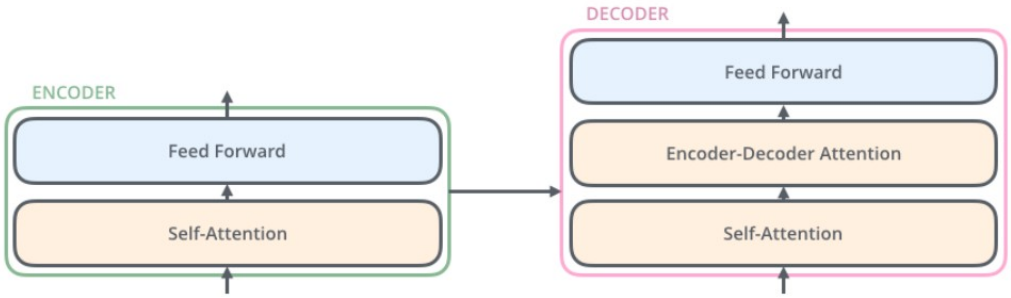
Transformers

- Transformers were one of the hottest new ideas in 2019. Transformers were introduced by the paper "Attention is all you need". This is a model architecture for the machine translation task which uses attention to outperform the previous SOTA, sequence-to-sequence models with attention, on many tasks.
- Transformers are a sequence-to-sequence model architecture: they still have a separate encoder and decoder component. The difference is in their use of an improved form of attention known as self-attention, which in addition to its expressive power has the computational advantage that it's expressable as a matrix operation.
- The core architecture consists of a stack of encoders fully connected to a stack of decoders. Each encoder consists of two blocks: a self-attention component, and a feed forward network. Each decoder consists of three blocks: a self-attention component, an encoder-decoder attention component, and a feed forward component.



Self-attention

- Vanilla attention is the use of a (typically linear) combination of previous hidden vectors in the input sequence (a sentence, in the case of a sequence-to-sequence model). This combination of hidden vectors is typically concatenated with the current word's vector and fed to the next layer of the model.
- Here's how attention is typically implemented in slightly more detail (in e.g. the PyTorch documentation's example). A zero matrix is created of sentence length. Each pass through the network populates a column in the matrix with a hidden vector for that word and word position. A hidden vector is generated for the next word. The prior hidden vector matrix is combined via a linear layer into a single combinatorial "prior context" vector, which is then concatenated to current word's hidden vector and passed to the next layer of the network.
- Self-attention is a new spin on the attention technique. Instead of looking at prior hidden vectors when considering a word embedding, self-attention is a weighted combination of all other word embeddings (including those that appear later in the sentence).
- A further innovation of the paper is multi-headed attention. The idea is that by repeating the self-attention mechanism multiple times the model can learn to

seperate different kinds of useful semantic information onto different channels (or heads).

Positional Encodings

- Self-attention is naively position-unaware, e.g. it encoded "The mouse ate the cat." and "The cat ate the mouse." the same way. Contrast this with vanilla attention and with recurrant neural network layers, which are both implicitly position-aware because they process embeddings sequentially.
- Adding this information into the model requires some form of positional encoding. In the paper transformer this is achieved by adding a positional vector to the word embedding prior to input to the self-attention layer. The position vector is a (sin, cos) structure: e.g. the first half of it is a bunch of sin operations on the current position, and the second half is a bunch of cos operations. Adding this to the embedding creates a learnable time dependency in the data; if this positional dependency is important to the task the model can use this information to learn it.

Installing Libraries

```
!pip install transformers
```

```
Collecting transformers
  Downloading transformers-4.18.0-py3-none-any.whl (4.0 MB)
ent already satisfied: tqdm>=4.27 in /usr/local/lib/python3.7/dist-
packages (from transformers) (4.63.0)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.7/dist-packages (from transformers) (1.21.5)
Requirement already satisfied: filelock in
/usr/local/lib/python3.7/dist-packages (from transformers) (3.6.0)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.7/dist-packages (from transformers)
(2019.12.20)
Requirement already satisfied: requests in
/usr/local/lib/python3.7/dist-packages (from transformers) (2.23.0)
Requirement already satisfied: importlib-metadata in
/usr/local/lib/python3.7/dist-packages (from transformers) (4.11.3)
Collecting huggingface-hub<1.0,>=0.1.0
  Downloading huggingface_hub-0.5.1-py3-none-any.whl (77 kB)
l>=5.1
  Downloading PyYAML-6.0-cp37-cp37m-
manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_12_x86_64.manylinux
2010_x86_64.whl (596 kB)
oses
  Downloading sacremoses-0.0.49-py3-none-any.whl (895 kB)
ent already satisfied: packaging>=20.0 in
/usr/local/lib/python3.7/dist-packages (from transformers) (21.3)
Collecting tokenizers!=0.11.3,<0.13,>=0.11.1
  Downloading tokenizers-0.11.6-cp37-cp37m-
manylinux_2_12_x86_64.manylinux2010_x86_64.whl (6.5 MB)
ent already satisfied: typing-extensions>=3.7.4.3 in
/usr/local/lib/python3.7/dist-packages (from huggingface-
```

```
hub<1.0,>=0.1.0->transformers) (3.10.0.2)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in
/usr/local/lib/python3.7/dist-packages (from packaging>=20.0-
>transformers) (3.0.7)
Requirement already satisfied: zipp>=0.5 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata-
>transformers) (3.7.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.7/dist-packages (from requests->transformers)
(2021.10.8)
Requirement already satisfied: urllib3!=1.25.0,!=1.25.1,<1.26,>=1.21.1
in /usr/local/lib/python3.7/dist-packages (from requests-
>transformers) (1.24.3)
Requirement already satisfied: chardet<4,>=3.0.2 in
/usr/local/lib/python3.7/dist-packages (from requests->transformers)
(3.0.4)
Requirement already satisfied: idna<3,>=2.5 in
/usr/local/lib/python3.7/dist-packages (from requests->transformers)
(2.10)
Requirement already satisfied: six in /usr/local/lib/python3.7/dist-
packages (from sacremoses->transformers) (1.15.0)
Requirement already satisfied: joblib in
/usr/local/lib/python3.7/dist-packages (from sacremoses->transformers)
(1.1.0)
Requirement already satisfied: click in /usr/local/lib/python3.7/dist-
packages (from sacremoses->transformers) (7.1.2)
Installing collected packages: pyyaml, tokenizers, sacremoses,
huggingface-hub, transformers
  Attempting uninstall: pyyaml
    Found existing installation: PyYAML 3.13
    Uninstalling PyYAML-3.13:
      Successfully uninstalled PyYAML-3.13
Successfully installed huggingface-hub-0.5.1 pyyaml-6.0 sacremoses-
0.0.49 tokenizers-0.11.6 transformers-4.18.0
```

```
!pip install datasets
```

```
Collecting datasets
  Downloading datasets-2.0.0-py3-none-any.whl (325 kB)
ent already satisfied: packaging in /usr/local/lib/python3.7/dist-
packages (from datasets) (21.3)
Collecting responses<0.19
  Downloading responses-0.18.0-py3-none-any.whl (38 kB)
Requirement already satisfied: pyarrow>=5.0.0 in
/usr/local/lib/python3.7/dist-packages (from datasets) (6.0.1)
Requirement already satisfied: pandas in
/usr/local/lib/python3.7/dist-packages (from datasets) (1.3.5)
Requirement already satisfied: dill in /usr/local/lib/python3.7/dist-
packages (from datasets) (0.3.4)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.7/dist-packages (from datasets) (1.21.5)
```

```
Requirement already satisfied: requests>=2.19.0 in
/usr/local/lib/python3.7/dist-packages (from datasets) (2.23.0)
Requirement already satisfied: huggingface-hub<1.0.0,>=0.1.0 in
/usr/local/lib/python3.7/dist-packages (from datasets) (0.5.1)
Requirement already satisfied: importlib-metadata in
/usr/local/lib/python3.7/dist-packages (from datasets) (4.11.3)
Collecting aiohttp
  Downloading aiohttp-3.8.1-cp37-cp37m-
manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_12_x86_64.manylinux
2010_x86_64.whl (1.1 MB)
ent already satisfied: multiprocess in /usr/local/lib/python3.7/dist-
packages (from datasets) (0.70.12.2)
Collecting xxhash
  Downloading xxhash-3.0.0-cp37-cp37m-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (212 kB)
ent already satisfied: tqdm>=4.62.1 in /usr/local/lib/python3.7/dist-
packages (from datasets) (4.63.0)
Requirement already satisfied: pyyaml in
/usr/local/lib/python3.7/dist-packages (from huggingface-
hub<1.0.0,>=0.1.0->datasets) (6.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in
/usr/local/lib/python3.7/dist-packages (from huggingface-
hub<1.0.0,>=0.1.0->datasets) (3.10.0.2)
Requirement already satisfied: filelock in
/usr/local/lib/python3.7/dist-packages (from huggingface-
hub<1.0.0,>=0.1.0->datasets) (3.6.0)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in
/usr/local/lib/python3.7/dist-packages (from packaging->datasets)
(3.0.7)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.7/dist-packages (from requests>=2.19.0-
>datasets) (2021.10.8)
Requirement already satisfied: idna<3,>=2.5 in
/usr/local/lib/python3.7/dist-packages (from requests>=2.19.0-
>datasets) (2.10)
Requirement already satisfied: chardet<4,>=3.0.2 in
/usr/local/lib/python3.7/dist-packages (from requests>=2.19.0-
>datasets) (3.0.4)
Requirement already satisfied: urllib3!=1.25.0,!=1.25.1,<1.26,>=1.21.1
in /usr/local/lib/python3.7/dist-packages (from requests>=2.19.0-
>datasets) (1.24.3)
Collecting urllib3!=1.25.0,!=1.25.1,<1.26,>=1.21.1
  Downloading urllib3-1.25.11-py2.py3-none-any.whl (127 kB)
ent already satisfied: charset-normalizer<3.0,>=2.0 in
/usr/local/lib/python3.7/dist-packages (from aiohttp->datasets)
(2.0.12)
Requirement already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.7/dist-packages (from aiohttp->datasets)
(21.4.0)
Collecting aiosignal>=1.1.2
```

```
  Downloading aiosignal-1.2.0-py3-none-any.whl (8.2 kB)
Collecting yarl<2.0,>=1.0
  Downloading yarl-1.7.2-cp37-cp37m-
manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_12_x86_64.manylinux
2010_x86_64.whl (271 kB)
ultidict<7.0,>=4.5
  Downloading multidict-6.0.2-cp37-cp37m-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (94 kB)
-
manylinux_2_5_x86_64.manylinux1_x86_64.manylinux_2_17_x86_64.manylinux
2014_x86_64.whl (144 kB)
eout<5.0,>=4.0.0a3
  Downloading async_timeout-4.0.2-py3-none-any.whl (5.8 kB)
Requirement already satisfied: zipp>=0.5 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata-
>datasets) (3.7.0)
Requirement already satisfied: python-dateutil>=2.7.3 in
/usr/local/lib/python3.7/dist-packages (from pandas->datasets) (2.8.2)
Requirement already satisfied: pytz>=2017.3 in
/usr/local/lib/python3.7/dist-packages (from pandas->datasets)
(2018.9)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.7/dist-packages (from python-dateutil>=2.7.3-
>pandas->datasets) (1.15.0)
Installing collected packages: multidict, frozenlist, yarl, urllib3,
asynctest, async-timeout, aiosignal, fsspec, aiohttp, xxhash,
responses, datasets
  Attempting uninstall: urllib3
    Found existing installation: urllib3 1.24.3
    Uninstalling urllib3-1.24.3:
      Successfully uninstalled urllib3-1.24.3
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
datascience 0.10.6 requires folium==0.2.1, but you have folium 0.8.3
which is incompatible.
Successfully installed aiohttp-3.8.1 aiosignal-1.2.0 async-timeout-
4.0.2 asynctest-0.13.0 datasets-2.0.0 frozenlist-1.3.0 fsspec-2022.3.0
multidict-6.0.2 responses-0.18.0 urllib3-1.25.11 xxhash-3.0.0 yarl-
1.7.2
```

```
from datasets import load_dataset
import tensorflow as tf
from transformers import AutoTokenizer,
TFAutoModelForSequenceClassification
from transformers import DataCollatorWithPadding
import tarfile
```

Loading the Dataset

```
compressed_file = '/content/drive/MyDrive/Colab Notebooks/extra
assignment/yelp_review_polarity_csv.tgz'
path = '/content/drive/MyDrive/Colab Notebooks/extra assignment/yelp'


def extract(data):
  tar = tarfile.open(data)
  tar.extractall(path)
  tar.close()
extract(compressed_file)

from google.colab import drive
drive.mount('/content/drive')

Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).

dataset_train = load_dataset('csv',
data_files='/content/drive/MyDrive/Colab Notebooks/extra
assignment/yelp/yelp_review_polarity_csv/train.csv',
column_names=['label','text'], split='train[:2%]',engine='python')

Using custom data configuration default-0c5a4e34cda4cb0c

Downloading and preparing dataset csv/default to
/root/.cache/huggingface/datasets/csv/default-0c5a4e34cda4cb0c/0.0.0/4
33e0ccc46f9880962cc2b12065189766fbb2bee57a221866138fb9203c83519...

{"version_major":2,"version_minor":0,"model_id":"8e45e886ef08433887d3b
d7ffc88e3e5"}

{"version_major":2,"version_minor":0,"model_id":"2a95c04580de421ebc8bc
7e01b738fcd"}

Dataset csv downloaded and prepared to
/root/.cache/huggingface/datasets/csv/default-0c5a4e34cda4cb0c/0.0.0/4
33e0ccc46f9880962cc2b12065189766fbb2bee57a221866138fb9203c83519.
Subsequent calls will reuse this data.

dataset_test = load_dataset('csv',
data_files={'test':'/content/drive/MyDrive/Colab Notebooks/extra
assignment/yelp/yelp_review_polarity_csv/test.csv'},
column_names=['label','text'],split='test[:10%]',engine='python')

Using custom data configuration default-12d3f80c7c18d366

Downloading and preparing dataset csv/default to
/root/.cache/huggingface/datasets/csv/default-
12d3f80c7c18d366/0.0.0/433e0ccc46f9880962cc2b12065189766fbb2bee57a2218
66138fb9203c83519...
```

{"version_major":2,"version_minor":0,"model_id":"01f33e3189254e48a663ac720a8f70e0"}

{"version_major":2,"version_minor":0,"model_id":"02d341c82fdd4877b26df2200f765cb6"}

```
Dataset csv downloaded and prepared to
/root/.cache/huggingface/datasets/csv/default-
12d3f80c7c18d366/0.0.0/433e0ccc46f9880962cc2b12065189766fbb2bee57a2218
66138fb9203c83519. Subsequent calls will reuse this data.
```

```
dataset_train
```

```
Dataset({
    features: ['label', 'text'],
    num_rows: 11200
})
```

```
dataset_test
```

```
Dataset({
    features: ['label', 'text'],
    num_rows: 3800
})
```

BERT

```
checkpoint_bert = 'distilbert-base-uncased'
tokenizer_bert = AutoTokenizer.from_pretrained(checkpoint_bert)
model_bert =
TFAutoModelForSequenceClassification.from_pretrained(checkpoint_bert,
num_labels=1)
```

{"version_major":2,"version_minor":0,"model_id":"1a29b422a6424391a1a0910638b8a891"}

{"version_major":2,"version_minor":0,"model_id":"c304134a9cf94d3795e58affac3956c2"}

{"version_major":2,"version_minor":0,"model_id":"38610ccc53e246c7b7cc2f6c0234cb5b"}

{"version_major":2,"version_minor":0,"model_id":"05e826b2411f4d529c3b42c89b2a222e"}

{"version_major":2,"version_minor":0,"model_id":"7de1cedc1e55434697403138f15b4b4a"}

```
Some layers from the model checkpoint at distilbert-base-uncased were
not used when initializing TFDistilBertForSequenceClassification:
['vocab_layer_norm', 'vocab_projector', 'vocab_transform',
'activation_13']
```

```
- This IS expected if you are initializing
TFDistilBertForSequenceClassification from the checkpoint of a model
trained on another task or with another architecture (e.g.
initializing a BertForSequenceClassification model from a
BertForPreTraining model).
- This IS NOT expected if you are initializing
TFDistilBertForSequenceClassification from the checkpoint of a model
that you expect to be exactly identical (initializing a
BertForSequenceClassification model from a
BertForSequenceClassification model).
Some layers of TFDistilBertForSequenceClassification were not
initialized from the model checkpoint at distilbert-base-uncased and
are newly initialized: ['dropout_19', 'classifier', 'pre_classifier']
You should probably TRAIN this model on a down-stream task to be able
to use it for predictions and inference.
```

```
model_bert.summary()
```

```
Model: "tf_distil_bert_for_sequence_classification"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 distilbert (TFDistilBertMai  multiple                 66362880
 nLayer)

 pre_classifier (Dense)      multiple                  590592

 classifier (Dense)          multiple                  769

 dropout_19 (Dropout)        multiple                  0

=================================================================
Total params: 66,954,241
Trainable params: 769
Non-trainable params: 66,953,472
_________________________________________________________________
```

```
for layer in model_bert.layers[:-2]:
 layer.trainable = False
```

```
def tokenize_bert(sample):
 return tokenizer_bert(sample['text'], truncation=True)
```

```
dataset_tokens_train_bert = dataset_train.map(tokenize_bert,
batched=True)
```


```
{"version_major":2,"version_minor":0,"model_id":"0bde48df2bd54b7b8e907
360b127cf60"}
```


```
dataset_tokens_test_bert = dataset_test.map(tokenize_bert,
batched=True)
```

{"version_major":2,"version_minor":0,"model_id":"6344405303b14cdba8a02d92ddedadbf"}

```
dataset_tokens_train_bert
```

```
Dataset({
    features: ['label', 'text', 'input_ids', 'attention_mask'],
    num_rows: 11200
})
```

```
dataset_tokens_test_bert
```

```
Dataset({
    features: ['label', 'text', 'input_ids', 'attention_mask'],
    num_rows: 3800
})
```

```
collate_fn_bert =
DataCollatorWithPadding(tokenizer=tokenizer_bert,return_tensors="tf")
```

```
train_data_bert = dataset_tokens_train_bert.to_tf_dataset(
 columns=['input_ids','token_type_ids','attention_mask'],
 label_cols=['label'],
 shuffle=True,
 collate_fn=collate_fn_bert,
 batch_size=32
)
```

```
test_data_bert = dataset_tokens_test_bert.to_tf_dataset(
 columns=['input_ids','token_type_ids','attention_mask'],
 label_cols=['label'],
 shuffle=False,
 collate_fn=collate_fn_bert,
 batch_size=32
)
```

```
model_bert.compile(optimizer='adam',
loss=tf.keras.losses.BinaryCrossentropy(from_logits=True),
metrics=['accuracy'])
```

```
tfb_call_bert =
tf.keras.callbacks.TensorBoard('/content/drive/MyDrive/Colab
Notebooks/extra assignment/yelp/tb_logs_bert',histogram_freq=1)
```

```
hist1 = model_bert.fit(train_data_bert, epochs=3,
callbacks=[tfb_call_bert])
```

```
Epoch 1/3
350/350 [==============================] - 482s 1s/step - loss: -
4.9134 - accuracy: 0.5278
Epoch 2/3
```

```
350/350 [==============================] - 475s 1s/step - loss: -
13.8797 - accuracy: 0.5278
Epoch 3/3
350/350 [==============================] - 475s 1s/step - loss: -
23.2644 - accuracy: 0.5278

model_bert.evaluate(test_data_bert)

119/119 [==============================] - 138s 1s/step - loss: -
27.0227 - accuracy: 0.5476

[-27.022666931152344, 0.547631561756134]
```

XLNET

```
checkpoint_xl = 'xlnet-base-cased'
tokenizer_xl = AutoTokenizer.from_pretrained(checkpoint_xl)
model_xl =
TFAutoModelForSequenceClassification.from_pretrained(checkpoint_xl,
num_labels=1)

{"version_major":2,"version_minor":0,"model_id":"363a3629d4fd4574b85bd
608a570d5e6"}

{"version_major":2,"version_minor":0,"model_id":"2b91d7e543184de394a5c
649a277dc91"}

{"version_major":2,"version_minor":0,"model_id":"230bffeee5ad48a8aa851
ba9d2632651"}

{"version_major":2,"version_minor":0,"model_id":"a086b2b34d024e85aa4da
c731f48c0c7"}

Some layers from the model checkpoint at xlnet-base-cased were not
used when initializing TFXLNetForSequenceClassification: ['lm_loss']
- This IS expected if you are initializing
TFXLNetForSequenceClassification from the checkpoint of a model
trained on another task or with another architecture (e.g.
initializing a BertForSequenceClassification model from a
BertForPreTraining model).
- This IS NOT expected if you are initializing
TFXLNetForSequenceClassification from the checkpoint of a model that
you expect to be exactly identical (initializing a
BertForSequenceClassification model from a
BertForSequenceClassification model).
Some layers of TFXLNetForSequenceClassification were not initialized
from the model checkpoint at xlnet-base-cased and are newly
initialized: ['sequence_summary', 'logits_proj']
You should probably TRAIN this model on a down-stream task to be able
to use it for predictions and inference.

model_xl.summary()
```

```
Model: "tfxl_net_for_sequence_classification"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 transformer (TFXLNetMainLay  multiple                 116718336
 er)

 sequence_summary (TFSequenc  multiple                 590592
 eSummary)

 logits_proj (Dense)         multiple                  769

=================================================================
Total params: 117,309,697
Trainable params: 117,309,697
Non-trainable params: 0
_________________________________________________________________
```

```
for layer in model_xl.layers[:-1]:
 layer.trainable = False
```

```
def tokenize_xl(sample):
 return tokenizer_xl(sample['text'], truncation=True)
```

```
dataset_tokens_train_xl = dataset_train.map(tokenize_xl, batched=True)
```


```
{"version_major":2,"version_minor":0,"model_id":"332d1431f9004d3e81bbc
7c88b5afe67"}
```


```
Asking to truncate to max_length but no maximum length is provided and
the model has no predefined maximum length. Default to no truncation.
```

```
dataset_tokens_test_xl = dataset_test.map(tokenize_xl, batched=True)
```


```
{"version_major":2,"version_minor":0,"model_id":"6a4f63158ab54b56bc30f
8d30793d529"}
```


```
dataset_tokens_train_xl
```

```
Dataset({
    features: ['label', 'text', 'input_ids', 'token_type_ids',
'attention_mask'],
    num_rows: 11200
})
```

```
dataset_tokens_test_xl
```

```
Dataset({
    features: ['label', 'text', 'input_ids', 'token_type_ids',
'attention_mask'],
    num_rows: 3800
})
```

```
collate_fn_xl =
DataCollatorWithPadding(tokenizer=tokenizer_xl,return_tensors="tf")

train_data_xl = dataset_tokens_train_xl.to_tf_dataset(
 columns=['input_ids','token_type_ids','attention_mask'],
 label_cols=['label'],
 shuffle=True,
 collate_fn=collate_fn_xl,
 batch_size=8
)

test_data_xl = dataset_tokens_test_xl.to_tf_dataset(
 columns=['input_ids','token_type_ids','attention_mask'],
 label_cols=['label'],
 shuffle=False,
 collate_fn=collate_fn_xl,
 batch_size=8
)

model_xl.compile(optimizer='adam',
loss=tf.keras.losses.BinaryCrossentropy(from_logits=True),
metrics=['accuracy'])

hist2 = model_xl.fit(train_data_xl, epochs=3)

Epoch 1/3
1400/1400 [==============================] - 1856s 1s/step - loss: -
68.4661 - accuracy: 0.5096
Epoch 2/3
1400/1400 [==============================] - 1870s 1s/step - loss: -
204.1882 - accuracy: 0.5113
Epoch 3/3
1400/1400 [==============================] - 1867s 1s/step - loss: -
340.6022 - accuracy: 0.5136

model_xl.evaluate(test_data_xl.take(100))

100/100 [==============================] - 110s 1s/step - loss: -
427.0682 - accuracy: 0.5163

[-427.06817626953125, 0.5162500143051147]
```

Conclusion

- Acquired knowledge about the self-attention, positional encodings and encoder-decoders of the transformer architecture.
- Implemented huggingface BERT and XLNet transformers on Tensorflow to carry out text classification on the yelp reviews dataset.